

Palagani Sai Ram Venkat

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PROFESSIONAL SUMMARY

Electronics Engineering graduate with an advanced certification in Embedded Systems. Well-versed in embedded systems development, including hardware design, firmware development, and real-time operating systems. Eager to apply theoretical knowledge and practical skills in a dynamic professional setting.

INTERNSHIP

Vector India

Oct '24 — Apr '25

- Trained at Vector India on advanced embedded systems, specializing in C programming, data structures, and bus protocols/IP networking. Gained hands-on project experience and actively participated in technical workshops to stay updated on emerging trends.

EDUCATION

Bachelor of Technology in Electronics and Communication Engineering, CMR College of Engineering and Technology (GPA: 6.2) 2020 — 2024

Intermediate in MPC, Narayana Junior College, Hyderabad (GPA: 8.3) 2018 — 2020

SSC, Sri Vivekananda Vidyalayam (GPA: 7.3) 2017 — 2018

CERTIFICATIONS

Accomplished a certification in Embedded Systems from Vector India.

PROJECTS

Smart Traffic Light Control System Using Arduino Mega

- This system manages traffic flow at intersections, aiming for smooth vehicle movement. This system dynamically modifies the timing of traffic light sequence according to the quantity of vehicles on each lane of a route.
- Emergency vehicles can be prioritized and can safely pass through the traffic signals with the help of Blynk IOT App.

SMS-Based Thermal Monitoring and Threshold Management

- Developed a real-time temperature monitoring system using LPC2148 microcontroller, LM35 sensor, and GSM module, enabling SMS-based alerts and remote threshold updates.
- Used Embedded C to program a microcontroller (LPC2148) for reading temperature and displaying it on an LCD.
- Allowed users to change temperature limits and get updates by sending SMS commands from their phone.

Banking Management System

- Implemented a command-line based banking application to simulate core functionalities such as account creation, deposit, withdrawal, money transfer, and transaction history tracking.
- Used file handling and structure pointers with singly linked lists (SLL) to manage dynamic data and ensure persistent storage across sessions.
- Developed a modular, menu-driven application with separate source files and a makefile for build automation, enhancing code maintainability.

TECHNICAL SKILL

Programming Languages C, C++, Embedded C, Basic of Python, Basics of Data Structure

Protocols UART, I2C, SPI, CAN, Basics of TCP/IP

Microcontrollers LPC2148, LPC2129

Hardware Interfacing LED, Seven-Segment Display, LCD, Keypad, ADC

Operating Systems Linux, Windows

Compilers GCC, Keil Debugger, Flash Magic, Proteus

Microsoft Tools Word, Excel, Power Point Presentation

PUBLICATIONS

Published a journal paper titled "Smart Traffic Light Control System Using Arduino Mega" in IJRASET.